COURSE KICK-OFF

Building end-to-end agentic applications on Databricks

12 October - 29 November 2026

7 weeks of hands-on LLMOps



Your teachers



Maria Vechtomova

Core instructor

Co-founder of Cauchy
Databricks MVP
12+ years in Data & AI



Basak Eskili

Core instructor

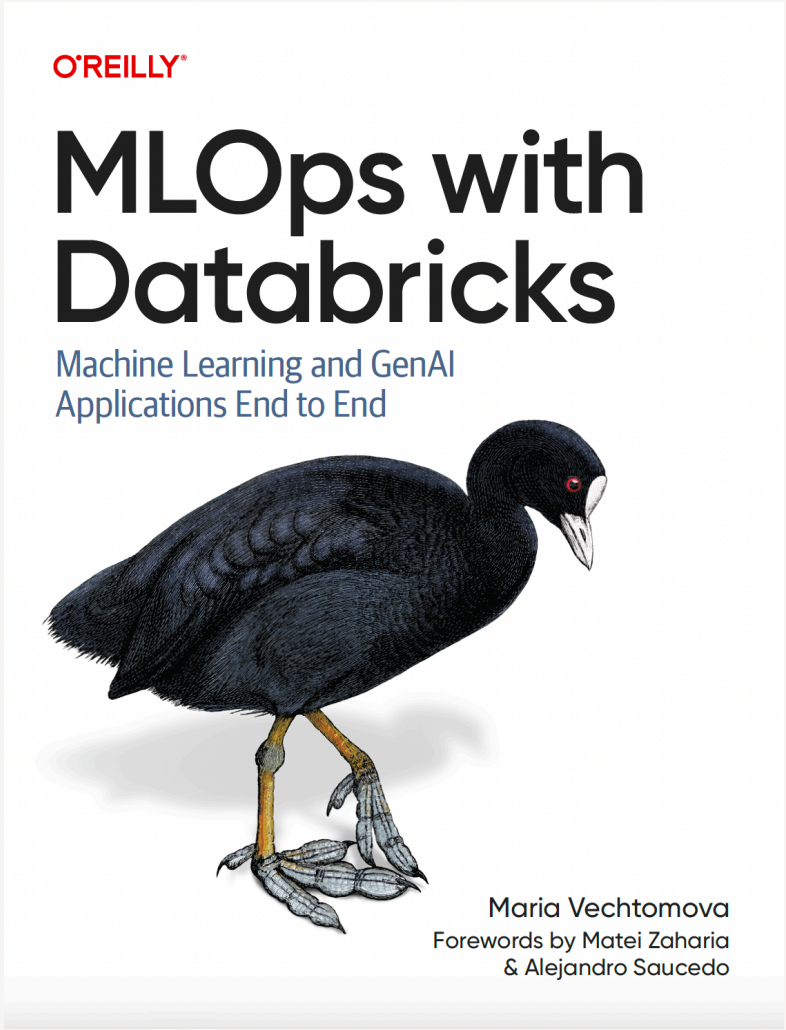
Senior ML Engineer at Booking.com
MLOps and production ML
10+ years in Data & AI



Victor Kuznetsov

Course assistant

Databricks Champion
Senior Data & AI Engineer at Cauchy
10+ years in Data & AI



MLOps with Databricks

Machine Learning and GenAI Applications End to End

Maria Vechtomova

About you

Bharadwaj Popuri

Harrisburg, Pennsylvania

About me

Engineering leader at the intersection of data platforms, machine learning, and life sciences.

Builds and modernizes clinical systems and data platforms for pharmaceutical companies and CROs.

Why I enrolled

Moving LLM projects into production in a regulated industry.

Staying hands-on with building, evaluating, deploying, and monitoring LLM applications on Databricks.

Fun fact	Drove and camped through the lower 48 US states
Hype song	Eye of the Tiger
Connect	bharadvajpopuri.com



Vivekkumar Muthukrishnan

London, England

About me

Staff / Lead Software Engineer working on data platforms.

Experience across network security, marketing, ecommerce, and travel.

Why I enrolled

Supporting a new LLMOps and MLOps initiative in his organization.

Expanding his platform skills and preparing for the transition.

Fun fact	Learning to play the guitar
Hype song	“Who Did You Think I Was” by John Mayer
Connect	LinkedIn

Your learning goals

Themes from your cohort submissions

Databricks capabilities

Explore the AI offering, new features, Vector Search, and agent workflows.

End-to-end LLMOps

Build, evaluate, deploy, monitor, and maintain LLM applications.

Production practices

Strengthen DAB development, governance, data quality, and observability.

Impact at work

Apply the learning to real use cases, guide architecture decisions, and share it with your teams.



In your words

What you want to take away from the course

"Gaining a better understanding of the OPS part of Agentic AI"

Pieter Holthuijsen

"I want to double-down on my knowledge on Databricks best practices such as DAB development."

Pablo Vizán Siso

"Connect LLMOps to Data Governance and Data Quality"

Lynn Overall

"Build hands-on experience with LLM projects across development, deployment, and evaluation, while helping define how we productionize Agentic AI solutions within the company"

Katy Chen



Practical details

How we learn

A shared example	We teach the concepts through an arXiv paper curator agent on Databricks.
Your own use case	Apply each week's concepts to an application you want to build.
Weekly PR reviews	Submit your work through GitHub pull requests for structured feedback.
Operations throughout	Agent design matters. Our focus is the system around the agent.



The weekly rhythm

WEDNESDAY SESSIONS

16:00-18:00

Amsterdam time

Recordings after each session

MONDAY MEETINGS

12 October

Course kick-off

23 November

Platform infrastructure introduction

Q&A throughout the week

Use Discord for Q&A throughout the week.
If we need extra Q&A, we can schedule a session.

Amsterdam: UTC+2 through 24 October, UTC+1 from 25 October. Calendar invites show your local time.

Curriculum

12 October - 29 November 2026

01 12-18 October

LLMOps Foundations & developing on Databricks

Mon 12: kick-off · Wed 14: teaching session

02 19-25 October

Data & knowledge processing

Wed 21 October

03 26 October - 1 November

MCP. Tool calling. Agent orchestration

Wed 28 October

04 2-8 November

MLflow for GenAI

Wed 4 November · Reflection due Thu 5

05 9-15 November

Deployment strategies for agents & DABs

Wed 11 November

06 16-22 November

Post-deployment: running AI in production

Wed 18 November

07 23-29 November

Platform infrastructure introduction

Mon 23 Nov

Demo day: show your project!

Wed 25 Nov

Your course access

Your starting point, unless we provided separate instructions to you



GitHub repository

Your dedicated repo

Accept the invitations for both the repository and the organization. Submit your project work as pull requests.

Repo + organization invite



Databricks access

Your course environment

You have access to our Databricks workspace and your own dashboard. We will walk through these together.

Workspace + personal dashboard



Discord

Private course channel

Ask questions, share progress, and get help throughout the week. We can schedule extra Q&A if needed.

Questions throughout the week

Course calendar, materials, and recordings: Maven

Your environment, live

Screen-sharing walkthrough · unless we provided separate instructions to you

01 Open Databricks and find your workspace

02 Open your personal dashboard and review what is available

03 Find your GitHub repository and the pull request workflow

04 Locate the calendar, materials, recordings, and Discord channel

Follow along with your own account. Flag any missing access during the walkthrough.



Materials and support

Reading on Maven

Slides, lecture transcriptions, and reading material.

Recordings on Maven

Session recordings available after each lecture.

Your progress

Use the course dashboard and Genie to track your performance.

Code feedback

Weekly GitHub PR reviews

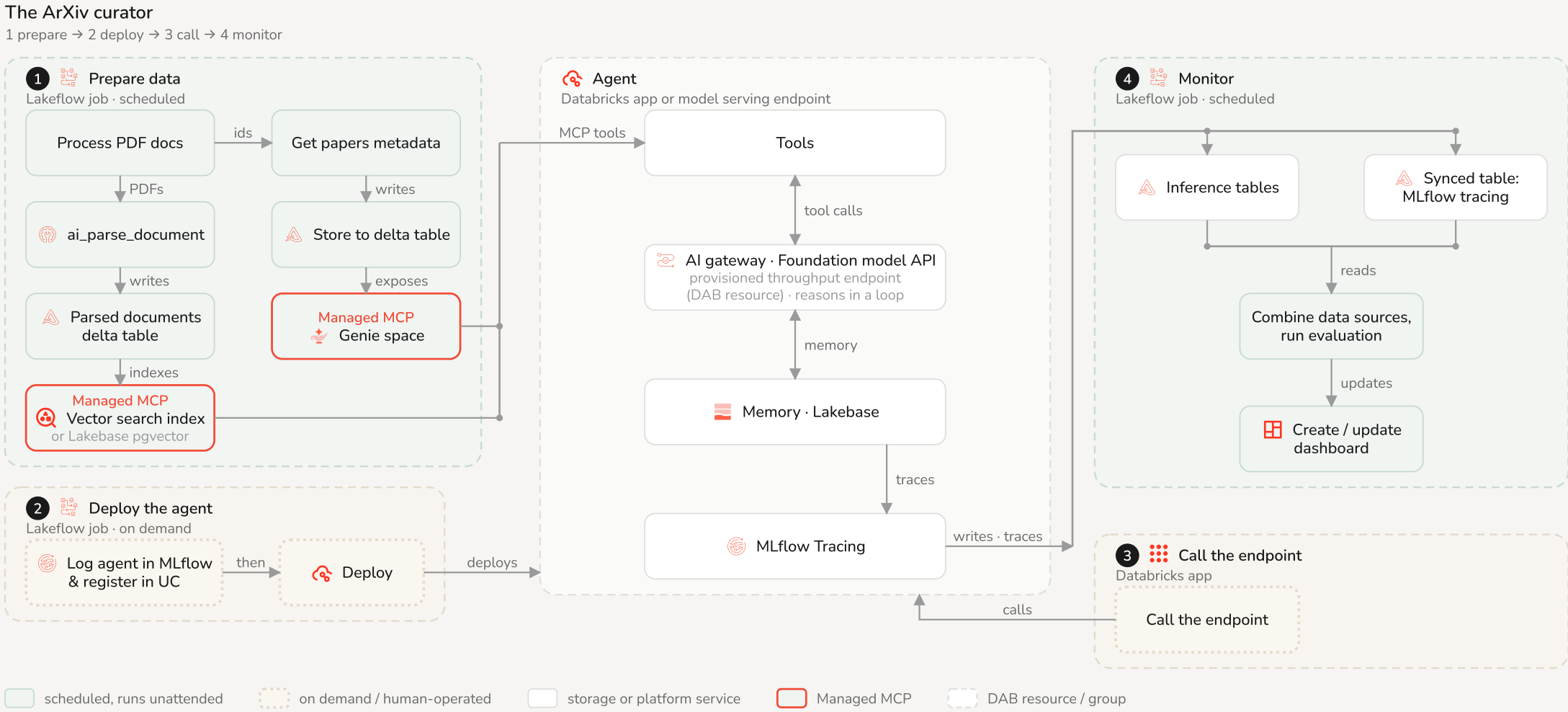
Q&A

Discord throughout the week
Q&As can be scheduled in case of demand.



The project we demonstrate

ArXiv curator



Your project

A use case you will develop throughout the course

A clear problem

Identify who will use the application and what it should help them do.

Data and tools

Choose the sources and capabilities your application needs.

Evidence of quality

Show traces and evaluation results alongside the application.

Demo day · 25 November

Demonstrate what you built and explain the operational decisions.

Before Wednesday

- Verify your Databricks access
- Accept your GitHub invitations
- Join Discord and check your calendar
- Bring an initial idea for your course project
- Fill in the form

  Unless we provided different instructions to you

Wednesday 14 October, 16:00 Amsterdam



MLOps vs LLMOps



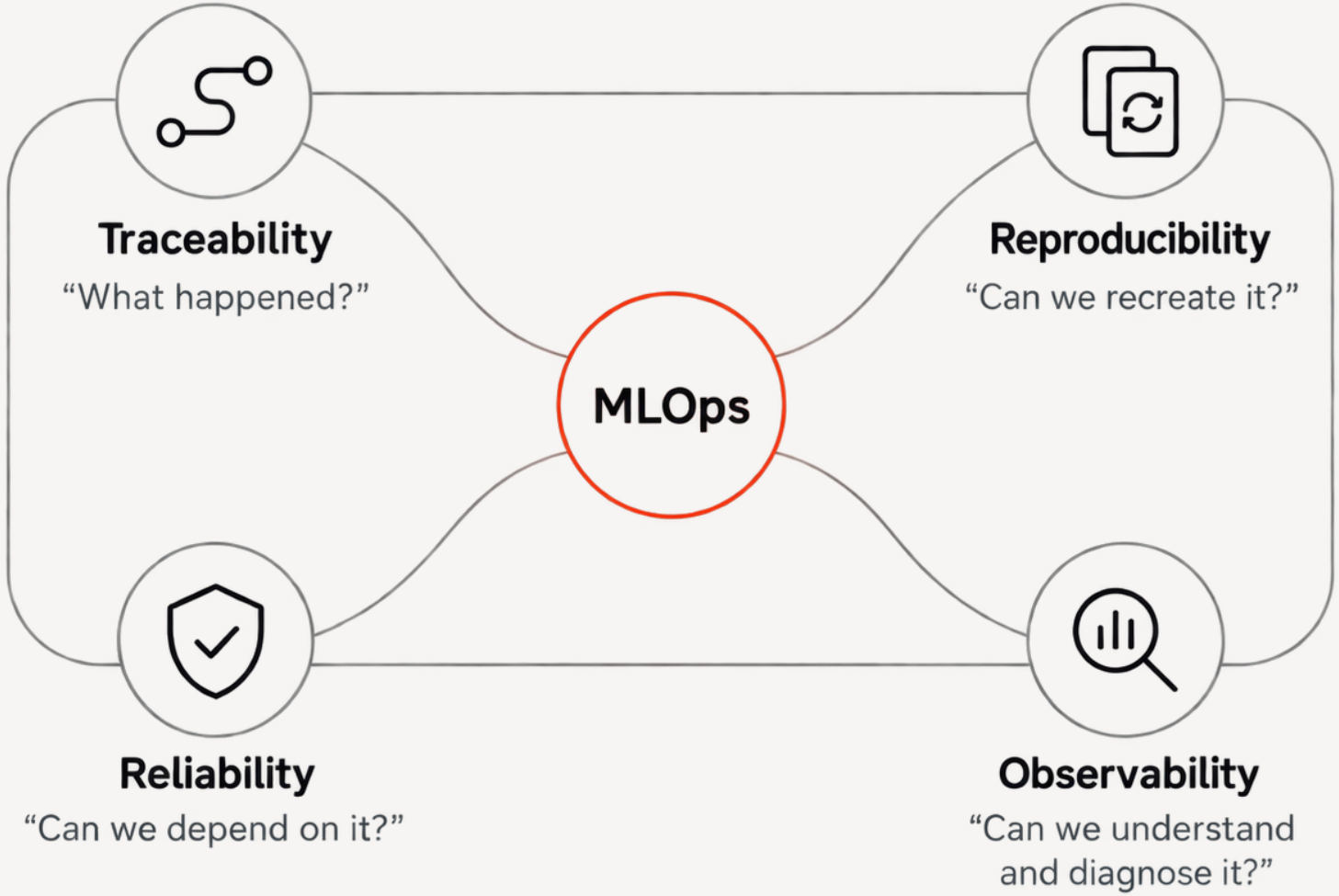
“MLOps is the set of practices for deploying and operating machine learning models reliably and efficiently.

LLMOps extends these practices to LLM workflows and agents.”

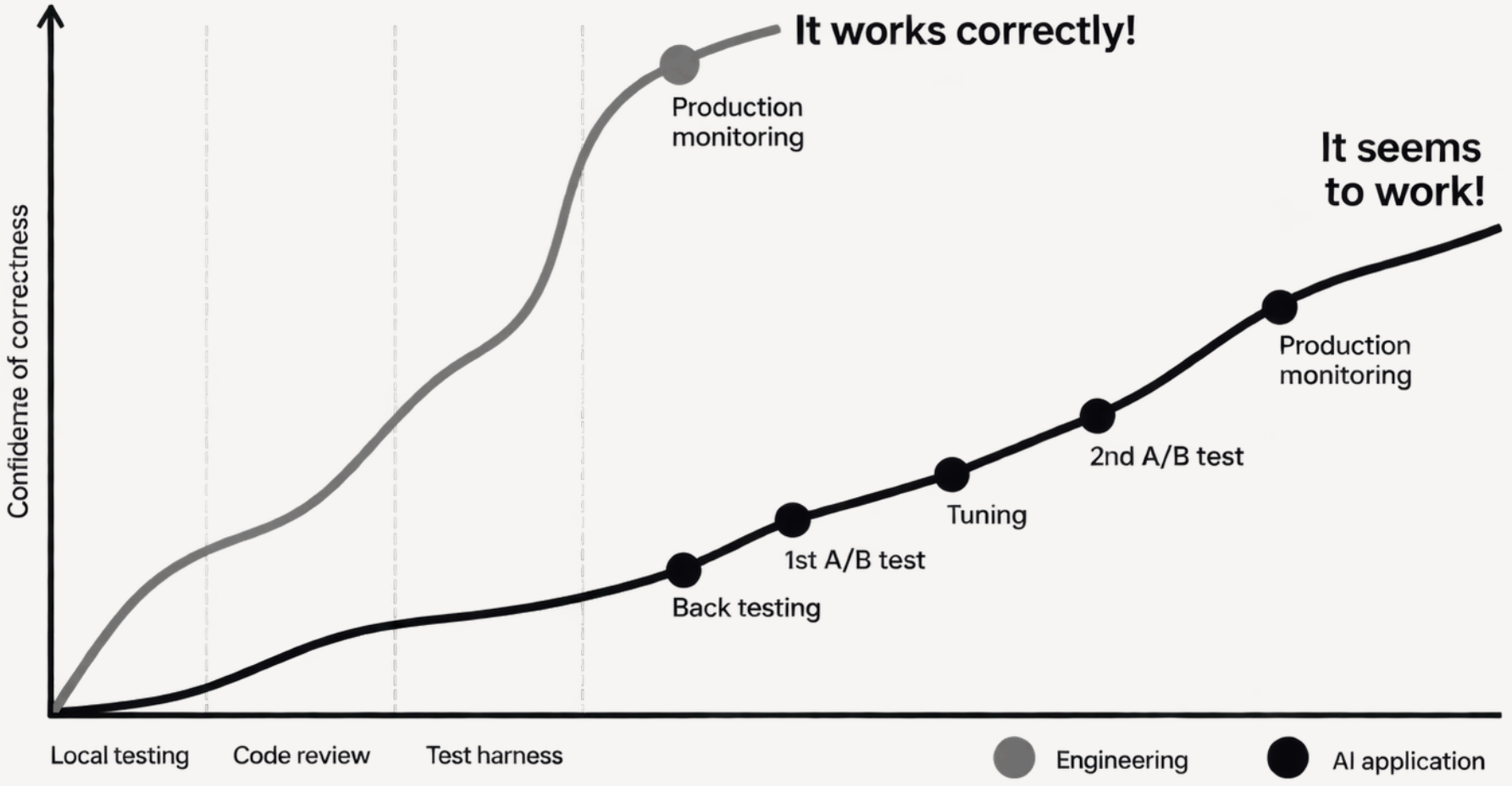


MLOps principles

These principles reinforce one another.



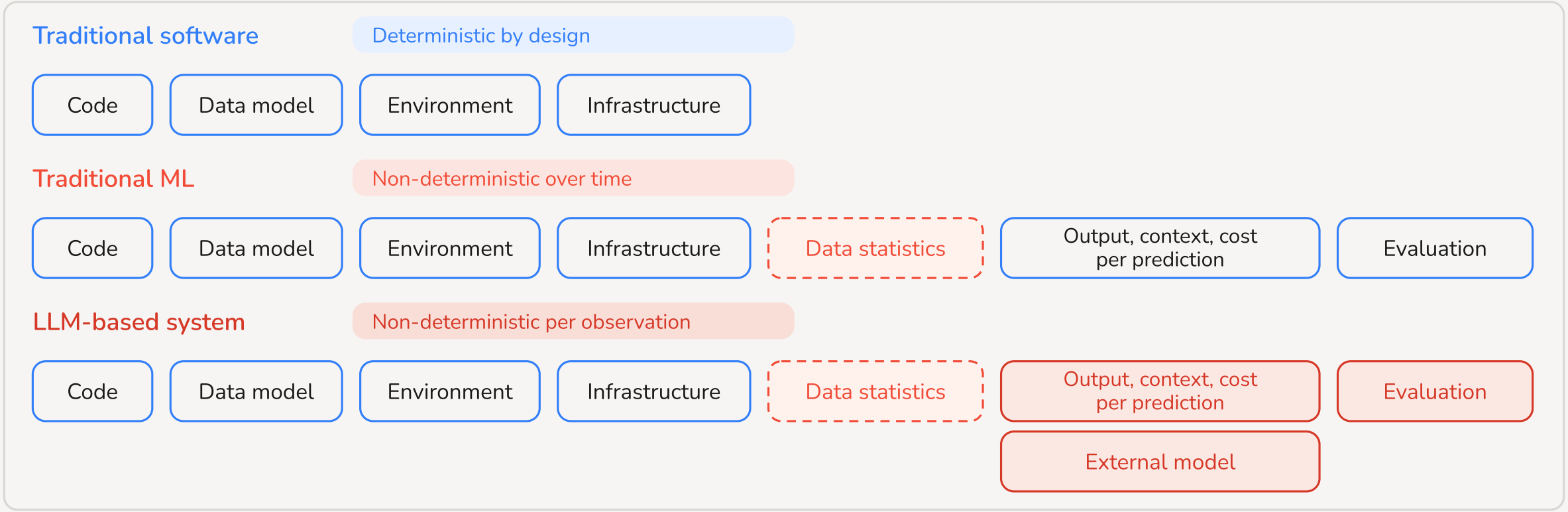
System lifecycle starts after deployment



The Complexity Ladder

Traditional software > Traditional ML > LLM-based system

Each step inherits the complexity of the previous one and adds a new source of non-determinism.

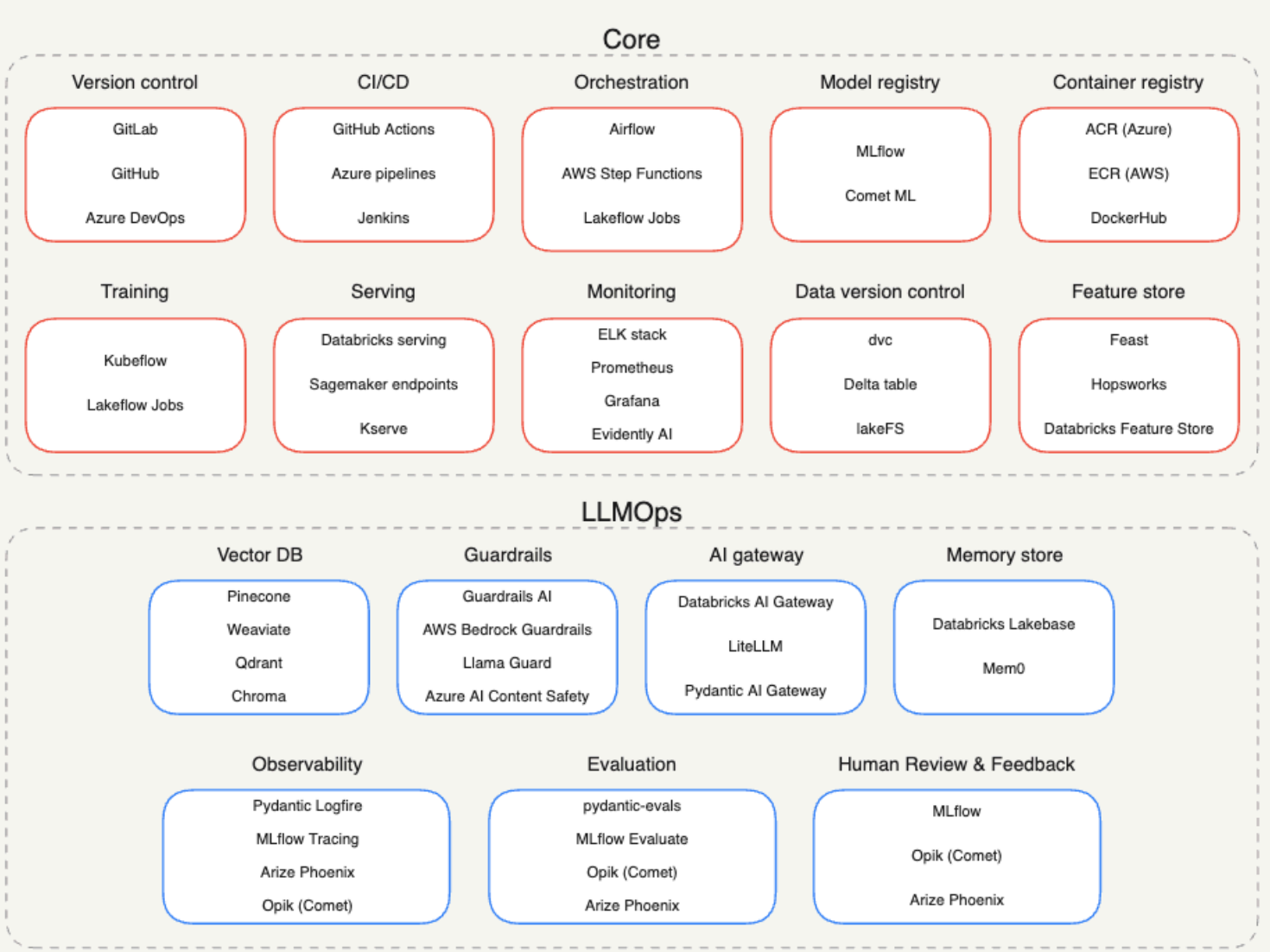


Key insight: A software system with identical properties produces identical output. An ML system cannot guarantee this.

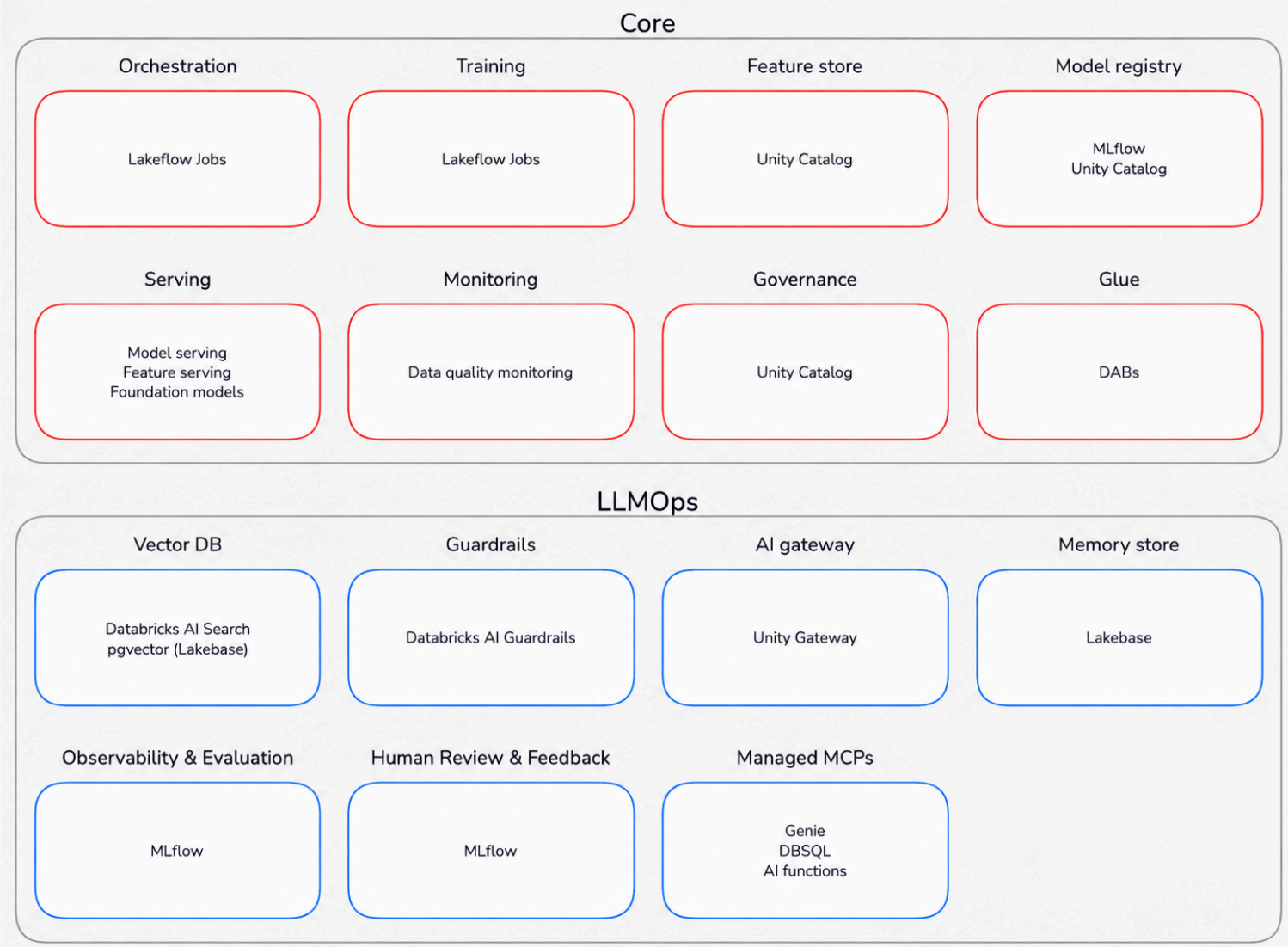
An LLM system adds an external model you do not own, plus non-deterministic output, retrieved context, cost, and evaluation.

The rungs are not clean steps: deep learning and NLP already blur them with GPU non-determinism and evaluation that was never fully objective. LLMs remove the last guarantees.

The wider tooling landscape



The Databricks ecosystem



Questions?

Course format, access, or your project idea

After today, continue the conversation on Discord.



Thank you

See you on Wednesday!

14 October · 16:00 Amsterdam

